

Ray Tsai

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EDUCATION

University of Oxford

Oxford, United Kingdom

M.Sc. in Mathematical Sciences, Grade: Distinction (75%), Rank: 13/77

Sep 2025 – Jun 2026

- **Dissertation:** *Induced Subgraphs in Ramsey Graphs*, advised by Prof. Alex Scott
- **Relevant coursework:** Probability on Graphs and Lattices, Probabilistic Combinatorics, Continuous Optimization, Theories of Deep Learning, Combinatorics

University of California San Diego

La Jolla, CA

B.Sc. in Mathematics-Computer Science (Honors with High Distinction), GPA: 3.93/4.0

Sep 2021 – Jun 2025

- **Honors Thesis:** *Double Turán Problems*, advised by Prof. Jacques Verstraete
- **Graduate coursework:** Statistical Learning, Combinatorics, Discrete Geometry, Complex Analysis

RESEARCH EXPERIENCE

UC San Diego, Department of Mathematics

La Jolla, CA

Undergraduate Research Assistant

Mar 2023 – Jun 2025

- Conducted independent research in extremal and probabilistic graph theory under Prof. Jacques Verstraete; proved new bounds on extremal problems involving Turán-type density conditions in graph families.
- Earned high distinction among the ~ 12 students annually admitted to the UCSD Math Honors Program.

UC San Diego, Department of CSE

La Jolla, CA

Undergraduate Research Assistant

Apr 2024 – Jun 2025

- Studied graduate-level computational complexity and learning theory directly under Prof. Russell Impagliazzo; covered PAC learning, boosting algorithms, and complexity-theoretic lower bounds.

WORK EXPERIENCE

Instructional Assistant

La Jolla, CA

UC San Diego, Department of CSE

Jan 2025 – Mar 2025

- Supported Design & Analysis of Algorithms (CSE 101) with 200+ students.

MSI

New Taipei City, Taiwan

Software Engineer Intern

Jul 2023 – Aug 2023

- Fine-tuned generative model pipeline in PyTorch; engineered automated installation and optimization backend, improving inference throughput by 80%.

Kook

San Diego, CA

Co-founder & Lead Software Engineer

Dec 2021 – Dec 2022

- Co-founded hardware-software startup; led 5-member engineering team, built full-stack mobile application in React-Native, Python, MySQL and embedded firmware in C++ for hardware control.

PROJECTS

Are Kalshi Prices Probabilities? | *Python, pandas, NumPy*

Oxford, 2026

- Built an end-to-end pipeline over the Kalshi public API testing whether prediction-market prices behave as calibrated probabilities, using Brier decomposition, logistic calibration fits, and event-clustered bootstrap confidence intervals.

Mathematical Analysis of Efficient Adversarial Robustness | *Python, PyTorch*

Oxford, 2026

- Unified FGSM-based training and Input Gradient Regularization via the Finlay-Oberman robustness bound, characterising catastrophic overfitting as exploitation of first-order approximation error.
- Confirmed on CIFAR-10 that smooth activations recover near-optimal robustness at a fraction of standard adversarial training cost.

AWARDS

Physical Sciences Dean's Undergraduate Award for Excellence

UC San Diego, 2025

- “Awarded to top undergraduates ($< 1\%$) in Mathematics, Chemistry & Biochemistry, and Physics for academic excellence and research promise.” (\$1000)